Project Title**: Reducing Hallucinations in Generative Text Models through Retrieval-Augmented and Self-Verification Decoding**

Details:
Team number:**18**

Project Domain: Large Language Models (LLMs) & RAG

Team Members: **Preksha Kamalesh** **PES2UG23CS902**
Navyashree SP **PES2UG23CS374**

Pranav Manoj **PES2UG23CS428**

Problem Statement and Abstract

Problem Statement:

- **The Challenge:** Large Language Models (LLMs) prioritize fluency over factuality, often generating "hallucinations"—plausible but incorrect information—even when provided with source text.
- **The Gap:** Standard Retrieval-Augmented Generation (RAG) retrieves evidence but lacks a robust mechanism to *verify* if the generated answer actually adheres to that evidence during the decoding process.

Abstract:

- We propose a hybrid generation framework to mitigate hallucinations in knowledge-intensive tasks.
- **Methodology:** The system integrates RAG with a Self-Verification Decoding mechanism. It first retrieves relevant documents, generates a response, and then performs claim-level verification against the retrieved evidence.
- **Outcome:** Unsupported or contradictory claims are filtered or regenerated, enhancing factual reliability without the computational cost of full model retraining.

Use Case of the Project

Primary Use Case: Automated Legal or Medical Query Answering

- **Scenario:** A user asks a specific question requiring high factual precision (e.g., "What are the side effects of Drug X?" or "Summarize the liability clause in Document Y").
- **Why this Project Matters:** In high-stakes domains, hallucinations are unacceptable. This system ensures answers are strictly grounded in the provided medical journals or legal contracts.

Secondary Use Case: Enterprise Knowledge Management

- **Scenario:** Internal chatbots answering employee queries based on company policy documents.
- **Benefit:** Prevents the AI from inventing policies that do not exist, ensuring compliance and accurate information retrieval.

Novelty and research gaps identified

1. Identified Research Gaps:

The "Context-Ignoring" Phenomenon: Standard RAG frameworks retrieve accurate documents, but LLMs often ignore this context in favor of their pre-trained (hallucinated) knowledge. There is a lack of effective mechanisms to force strict adherence to retrieved context during generation.

Lack of Granular Correction: Most self-correction methods regenerate the entire response if an error is found, which is inefficient. There is a gap in methods that perform claim-level editing (fixing only the specific wrong fact).

2. Novelty of Proposed Work:

Hybrid Verification Loop: We introduce a post-generation "Self-Verification" step that acts as a quality gate, using the model's own reasoning capabilities to cross-check generated claims against the retrieved evidence before showing the answer to the user.

Surgical Correction: The system identifies and filters specific unsupported claims rather than discarding the whole response, preserving the fluency of the valid parts of the answer.

Validation metrics

We will compare our model against baseline 1 (Vanilla LLM) and baseline 2 (Standard RAG) using the following metrics

FactScore (Precision): Measures the percentage of generated atomic claims supported by the source.

- **Goal: Beat the Standard RAG FactScore (typically ~60-70% depending on complexity) by achieving >80% precision.**

Hallucination Rate (NLI Scores): Using Natural Language Inference to classify sentences as "Entailment" (True) or "Contradiction" (Hallucination).

- **Goal: Reduce the "Contradiction" rate by at least 15-20% compared to the baseline.**

Inference Time Overhead: Since verification adds steps, the metric to watch is "Time per Token."

- **Goal: Keep the latency increase within acceptable limits (e.g., <1.5x of standard generation) to prove usability.**

Literature survey

Paper Title	Authors / Conference or Journal	Methodology Used	Results	Limitations
Towards Mitigating Hallucination in Large Language Models via Self-Reflection	Ziwei Ji, Tiezheng Yu, Yan Xu, Nayeon Lee, Etsuko Ishii, Pascale Fung – Findings of EMNLP 2023	Proposed an interactive self-reflection framework for LLMs. The method generates background knowledge, evaluates factuality using entailment checks, and iteratively refines answers until consistency with evidence is achieved. Evaluated using medical QA datasets such as PubMedQA and MedQuAD.	Experiments showed improved factual consistency and reduced hallucination compared to baseline LLM outputs. The iterative feedback loop enhanced accuracy and reliability in generative QA systems.	Iterative reasoning increases computational cost and latency. The approach mainly evaluated medical QA tasks, so generalization to other domains needs further testing.
Context Tuning for Retrieval Augmented Generation	Raviteja Anantha, Tharun Bethi, Danil Vodianik, Srinivas Chappidi – Apple Research / arXiv 2023	Introduces Context Tuning, a system that retrieves contextual information before tool retrieval in RAG pipelines. Uses semantic search, Chain-of-Thought augmentation, and ranking models such as LambdaMART with Reciprocal Rank Fusion for better context retrieval.	Achieved 3.5× improvement in Recall@K for context retrieval and 1.5× improvement for tool retrieval, along with 11.6% improvement in planner accuracy in LLM-based systems.	Requires additional retrieval models and ranking mechanisms, increasing system complexity. Performance depends heavily on quality of contextual data and retrieval features.
SED: Self-Evaluation Decoding Enhances Large Language Models for Better Generation	Ziqin Luo, Haixia Han, Haokun Zhao, Guochao Jiang, Chengyu Du, Tingyun Li, Jiaqing Liang, Deqing Yang, Yanghua Xiao – arXiv 2024	Proposed Self-Evaluation Decoding (SED). During text generation, the model detects uncertain tokens (“chaotic points”), speculates multiple candidate tokens, evaluates generated responses, and selects the best token using combined probability and evaluation scores.	Demonstrated improved generation quality and accuracy across tasks compared to standard decoding methods like greedy search or beam search. The approach reduces error propagation during generation.	Adds extra decoding steps, increasing inference time. Requires additional evaluation capability in the model and may not scale efficiently for very long outputs.

Literature survey

Chain-of-Verification Reduces Hallucination in Large Language Models	Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, Jason Weston – Meta AI, ETH Zurich – arXiv 2023	Proposes Chain-of-Verification (CoVe) to reduce hallucinations in LLMs. The method follows four steps: Generate a baseline response Create verification questions to check factual claims Answer the verification questions independently Generate a corrected final response using the verified information.	Experiments on datasets such as Wikidata, Wiki-Category, MultiSpanQA, and biography generation show that CoVe significantly reduces hallucinated content and improves factual accuracy. The FactScore improved from 55.9 to 71.4 compared to the baseline model.	The method increases computational cost and inference time due to multiple verification steps. It also cannot completely eliminate hallucinations and still depends on the knowledge capability of the base language model.
Corrective Retrieval Augmented Generation (CRAG)	Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, Zhen-Hua Ling – arXiv 2024	Proposes CRAG, a framework that improves RAG by evaluating retrieved documents before generation. A retrieval evaluator classifies results as Correct, Incorrect, or Ambiguous. Irrelevant documents trigger web search, while relevant documents are refined using knowledge filtering.	CRAG improves factual accuracy and generation quality compared to standard RAG and Self-RAG. Shows better performance across datasets like PopQA, Biography, PubHealth, and ARC-Challenge	Requires an additional retrieval evaluator model and extra processing steps, increasing system complexity and inference time.
RAGAS: Automated Evaluation of Retrieval Augmented Generation	Shahul Es, Jithin James, Luis Espinosa-Anke, Steven Schockaert – arXiv 2023	Proposes RAGAS, a framework for automated evaluation of RAG systems without requiring reference answers. It measures three key aspects: Faithfulness (whether the answer is supported by retrieved context), Answer Relevance (how well the answer addresses the question), and Context Relevance (quality of retrieved documents). The framework uses LLM prompts to compute these metrics automatically.	Experiments using the WikiEval dataset show that RAGAS metrics have high agreement with human judgments, achieving strong accuracy in evaluating faithfulness, answer relevance, and context relevance compared to baseline evaluation methods.	Evaluation still relies on LLM prompts and may be affected by prompt design. The framework can struggle with evaluating very long contexts or subtle differences in answers.

"RAG-HAT: A Hallucination-Aware Tuning Pipeline for LLM in Retrieval-Augmented Generation"	Juntong Song, Xingguang Wang, Juno Zhu, Yuanhao Wu, Xuxin Cheng, Randy Zhong, Cheng Niu – EMNLP Industry Track 2024	Proposes a Hallucination-Aware Tuning (HAT) pipeline combining detection and correction. A hallucination detection model (fine-tuned Llama-3-8B) is trained to generate detection labels and descriptions of hallucinations at both answer and sentence levels. GPT-4 Turbo is then used to correct the detected hallucinations. The corrected outputs are paired with original outputs to create a preference dataset for Direct Preference Optimization (DPO) fine-tuning of the base LLM. Evaluated on RAGTruth and WebGLM datasets.	Detection model achieved best-in-class performance on RAGTruth test set, outperforming even a fine-tuned Llama-2-13B baseline in F1 score. DPO-fine-tuned model demonstrated measurably reduced hallucination rates on a 1,000-example WebGLM set and 450-example RAGTruth set. Improved answer quality across both factual accuracy and fluency.	Requires full DPO fine-tuning — computationally expensive and not inference-time adaptable. Relies on GPT-4 Turbo for correction, introducing cost and API dependency. Detection model cannot fairly benchmark the hallucination rate of the regeneration approach. Whole-response correction rather than surgical claim-level editing, reducing efficiency for partially correct outputs.
MEGA-RAG: A Retrieval-Augmented Generation Framework with Multi-Evidence Guided Answer Refinement	PMC / Frontiers in Public Health – 2025	Introduces a multi-stage RAG framework combining three retrieval strategies: dense retrieval (FAISS), keyword-based retrieval (BM25), and biomedical knowledge graphs (KGs). A cross-encoder reranker ensures semantic relevance of retrieved evidence. A discrepancy-aware refinement module reconciles conflicts across evidence sources before generation to enhance factual accuracy. Benchmarked against PubMedBERT, PubMedGPT, standard LLM, and standard RAG baselines on public health QA.	Reduced hallucination rates by over 40% compared to vanilla RAG and standalone LLM baselines. Achieved highest scores across factual accuracy and evidence alignment metrics. Demonstrated strong performance in medical QA for clinical communication, health education, and evidence-based policy tasks.	Multi-stage design with multiple retrieval and LLM calls introduces substantial computational overhead. No detailed efficiency benchmark was provided — latency trade-offs are unquantified. Overhead from multi-source retrieval and reranking, though parallelizable, may hinder real-time applications. Primarily validated in the biomedical/public health domain; generalization to other domains untested.
ReDeEP: Detecting Hallucination in Retrieval-Augmented Generation via Mechanistic Interpretability	Anonymous Authors – OpenReview / NeurIPS Workshop 2024	Investigates internal LLM mechanisms behind hallucinations in RAG settings using mechanistic interpretability. Identifies two key components: Knowledge FFNs (feed-forward networks that store parametric knowledge) and Copying Heads (attention heads responsible for integrating retrieved external content). Proposes ReDeEP, a detection framework that decouples and scores the competition between parametric and retrieved knowledge at token generation time to flag hallucinations.	Hallucinations detected when Knowledge FFNs over-emphasize parametric memory while Copying Heads fail to retain external retrieved content. ReDeEP outperforms prior hallucination detection methods by accurately disentangling the dual knowledge sources within RAG pipelines. Provides interpretable, token-level hallucination signals.	Mechanistic interpretability approach requires white-box access to model internals — not applicable to closed-source or API-only LLMs. Computationally heavier than simple output-based classifiers. Detection-focused; does not provide a built-in correction or regeneration mechanism. Evaluated on a limited set of model architectures; broader generalizability is unclear.
Attention is all you need	Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, Illia Polosukhin – NeurIPS 2017	Introduces the Transformer architecture, eliminating recurrence and convolution entirely in favor of a self-attention mechanism. Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions. Positional encodings replace sequence-order signals previously provided by RNNs/LSTMs. Encoder-decoder structure used for sequence-to-sequence tasks. Trained on WMT 2014 English-German and English-French translation benchmarks.	Achieved state-of-the-art BLEU scores on WMT 2014: 28.4 on English-to-German and 41.0 on English-to-French, surpassing all prior models including ensembles. Training was significantly faster than RNN/CNN baselines due to full parallelization — the big model trained in 3.5 days on 8 P100 GPUs. Became the foundational architecture behind all modern LLMs including BERT, GPT, T5, and LLaMA.	Self-attention scales quadratically with sequence length ($O(n^2)$), making it expensive for very long inputs. No inherent positional awareness — requires explicit positional encodings which can be suboptimal. Lacks memory across conversations — no built-in mechanism for long-context retention. Highly data-hungry; requires large corpora and compute to generalize well. Prone to hallucinations when used generatively, as it optimizes for fluency over factual grounding — a core problem this project directly addresses.